

Introduction Computer Vision (CS641)

Lecture02 - Working with Images

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Image basics

Image types

- Binary
- Gray Scale
- Color

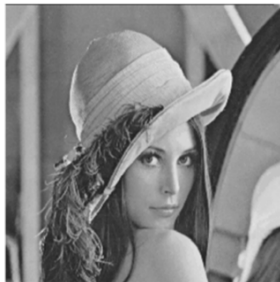


Figure: Image samples. Binary, Gray scale and color.

Image basics

Binary Images

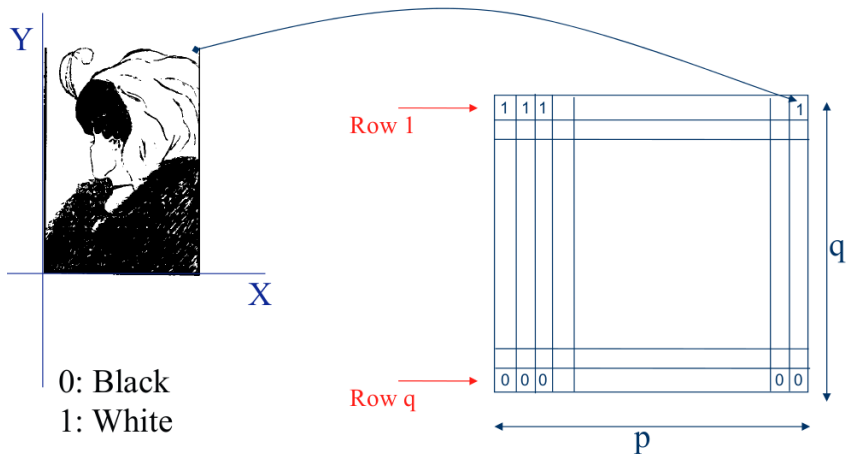


Figure: Binary Images.

Image basics

Gray Level Images

Grayscale is a range of shades of gray from 0 to 255 without apparent color.

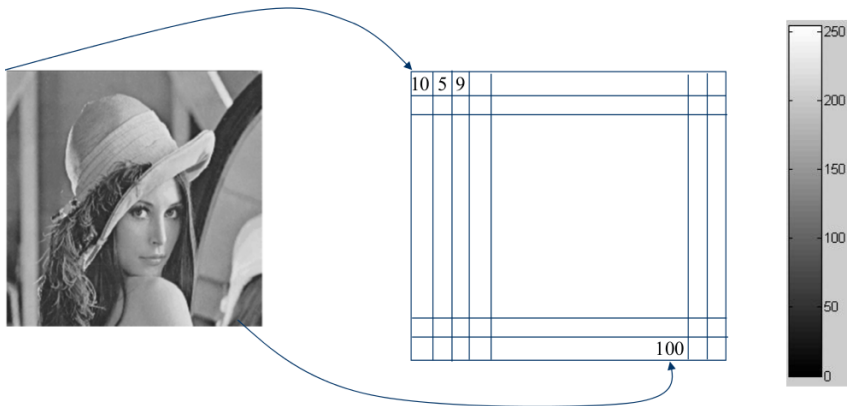


Figure: Gray Level Images.

Image basics

Gray Level Images

Adding a red color channel to the grayscale image.

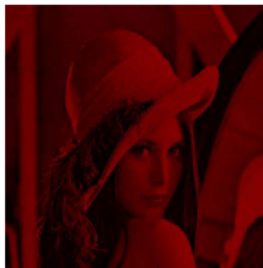


Figure: Gray Level Images with color channel.

Image basics

Color Images

Color images consists of red, green and blue channels.



Figure: Color Images.

Image Histogram

- An **image histogram** is a graphical representation of the number of pixels in an image as a function of their intensity.
- Histograms are made up of bins, each bin representing a certain intensity value range.

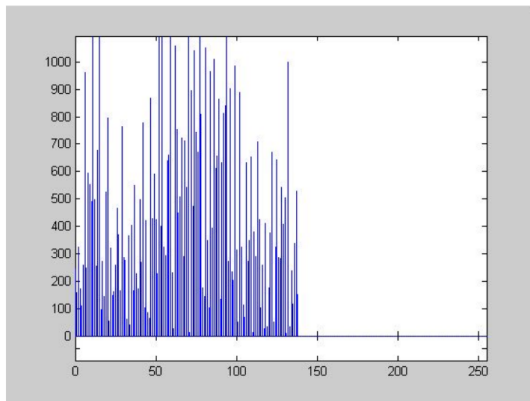
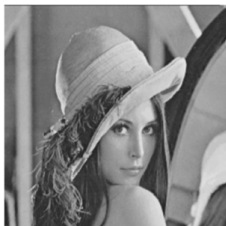


Figure: Image histogram

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Image Noise

The **noise** added to the image changes the actual pixel intensity value.
The reasons of noise in image

- Light variations
- Camera electronics
- Surface reflectance
- Lens

Image noise

- $I(x, y)$: the true pixel values
- $n(x, y)$: the noise at pixel (x, y)

$$\hat{I}(x, y) = I(x, y) + n(x, y)$$



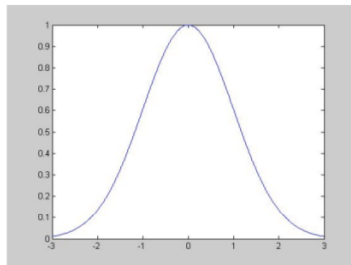
Figure: Noise added to the image.

Image noise

Gaussian noise

A function that represents the distribution of a variable (random variables) as a symmetrical bell-shaped graph.

$$n(x, y) = e^{\frac{-n^2}{2\sigma^2}}$$



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A sample program

```
#include "opencv2/core/core.hpp"
#include "opencv2/highgui/highgui.hpp"
#include <iostream>

using namespace cv;
using namespace std;

int main(int argc, const char *argv[])
{
    Mat image = imread ("lena.jpg", 1);

    if (! image.data )
    {
        cout << "image not found" << endl;
        return -1;
    }

    namedWindow("Window", 1);

    imshow("Window", image);
    waitKey(0);

    return 0;
}
```

- Grayscale image

```
cvtColor(src, dst, CV_BGR2GRAY);
```

- Thresholding to binary

```
#include "opencv2/imgproc/imgproc.hpp"
```

```
threshold (src_gray, dst_bin, 120, 255, THRESH_BINARY);
```

OpenCV functions

filter

- Create your own box filter

```
kernel = Mat::ones( 3, 3, CV_32F ) / 9;
```

```
filter2D(src, dst, ddepth , kernel, anchor, delta,  
BORDER_DEFAULT );
```

OpenCV functions

Smoothing images

- Block filter

blur(src, dst, Size(i, i), Point(-1,-1));

- Gaussian filter

GaussianBlur(src, dst, Size(i, i), 0, 0);

- Median filter

medianBlur (src, dst, i);

Next class - Edge Detection